MODULE-2

PROGRAMMING PRACTICE USING JAVA

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SECTION: - A8

YEAR: - 3rd

TOPICS COVERED: -

- **LINEAR SEARCH**
- **BINARY SEARCH**
- **BUBBLE SORT**
- **INSERTION SORT**
- **SELECTION SORT**

MODULE - 2

1. Fraud Transaction Detection (Linear Search)

Code:

```
import java.util.Scanner;
public class FraudTransaction{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for(int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }
        int target = sc.nextInt();
        boolean found = false;
        for(int i=0; i<n; i++){
            if(arr[i] == target){
                found = true;
                break;
            }
        }
        if(found){
            System.out.println("Fraud Transaction
                                    Found");
        }else{
            System.out.println(" Transaction Not
                                    Found");
        }
        sc.close();
    }
}
```

```java
package module;
import java.util.Scanner;
public class FraudTransaction {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int target = sc.nextInt();
        boolean found = false;
        for (int i = 0; i < n; i++) {
            if (arr[i] == target) {
                found = true;
                break;
            }
        }
        if (found) {
            System.out.println("Fraud Transaction Found");
        } else {
            System.out.println("Transaction Not Found");
        }
        sc.close();
    }
}
```

@ Javadoc | Declaration | Console

<terminated> FraudTransaction [Java Application] C:\Users\mahi\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-06

```
5
1201 1305 1450 1408 1502
1450
Fraud Transaction Found
```

2. Server Log Search (Binary Search)

Code:
```
import java.util.Scanner;
public class ServerlogSearch {
    public static void main(String[] args){
        Scanner sc = new Scanner (System.in);
        int n = sc.nextInt();
        int arr [] = new int [n];
        for (int i = 0; i<n; i++){
            arr[i] = sc.nextInt();
        }
        int target = sc.nextInt();
        int low = 0;
        int high = n-1;
        boolean found = false;
        while (low <= high) {
            int mid = (low+ high)/2;
            if (arr[mid] == target {
                found = true;
                break;
            }else if (arr [mid]< target){
                low = mid +1;
            }else{
                high = mid -1;
            }
        }
        if (found){
            System.out.println("Log Found");
        }else{
            System.out.println("Log Not Found")
        }
        sc.close();
    }
}
```

3.

Code

```java
package module;
import java.util.Scanner;
public class ServerLogSearch {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int target = sc.nextInt();
        int low = 0;
        int high = n - 1;
        boolean found = false;
        while (low <= high) {
            int mid = (low + high) / 2;
            if (arr[mid] == target) {
                found = true;
                break;
            } else if (arr[mid] < target) {
                low = mid + 1;
            } else {
                high = mid - 1;
            }
        }
        if (found) {
            System.out.println("Log Found");
        } else {
            System.out.println("Log Not Found");
        }
        sc.close();
    }
}
```

Javadoc Declaration Console

```
<terminated> ServerLogSearch [Java Application] C:\Users\mahi\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-0
7
1001 1005 1010 1015 1020 1030 1040
1015
Log Found
```

3. Sort Employee Salaries (Bubble Sort)

Code:

```
import java.util.Scanner;
public class BubbleSortSalary {
    public static void main(String [] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int [n];
        for(int i =0; i<n ; i++){
            arr[i] = sc.nextInt();
        }
        for( int i=0 ; i<n-1; i++){
            for(int j =0 ; j< n-i-1 ; j++){
                if(arr[j] > arr [j+1]){
                    int temp = arr[j];
                    arr[j] = arr [j+1];
                    arr [j+1] = temp;
                }
            }
        }
        for (int i = 0; i<n ; i++){
            System.out.print (arr[i] + " ");
        }
        sc.close();
    }
}
```

```java
package module;
import java.util.Scanner;
public class BubbleSortSalary {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        for (int i = 0; i < n - 1; i++) {
            for (int j = 0; j < n - i - 1; j++) {
                if (arr[j] > arr[j + 1]) {
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                }
            }
        }
        for (int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }
        sc.close();
    }
}
```

@ Javadoc   Declaration   Console

```
<terminated> BubbleSortSalary [Java Application] C:\Users\mahi\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-
5
45000 32000 55000 28000 60000
28000 32000 45000 55000 60000
```

Date
DELTA Pg No.

4. Sort Website Response Time (Insertion Sort)

Code:-
```
import java.util.Scanner;
public class InsertionSortResponse {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for(int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }
        for (int i = 1; i<n; i++){
            intkey = arr[i];
            int j = i-1;
            while (j >= 0 && arr[j] > key){
                arr[j+1] = arr[j];
                j--;
            }
            arr[j+1] = key;
        }
        for(int i=0; i<n; i++){
            System.out.print(arr[i]+" ");
        }
        sc.close();
    }
}
```

```java
package module;
import java.util.Scanner;
public class InsertionSortResponse {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        for (int i = 1; i < n; i++) {
            int key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j] > key) {
                arr[j + 1] = arr[j];
                j--;
            }
            arr[j + 1] = key;
        }
        for (int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }
        sc.close();
    }
}
```

@ Javadoc | Declaration | Console

```
<terminated> InsertionSortResponse [Java Application] C:\Users\mahi\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-
5
250 120 320 180 100
100 120 180 250 320
```

Date
DELTA Pg No.

5. Prioritize Bug Reports (Selection Sort)

Code :-

```
import java.util.Scanner;
public class SelectionSortBug{
   public static void main(String[] args){
      Scanner sc = new Scanner(System.in);
      int n = sc.nextInt();
      int arr[] = new int[n];
      for(int i = 0; i<n; i++){
         int min = i;
         for (int j = i+1; j<n; j++){
            if(arr[j]< arr[min]){
               min = j;
            }
         }
         int temp = arr[i];
         arr[i] = arr[min];
         arr[min] = temp;
      }
      for(int i = 0; i< n; i++){
         System.out.print(arr[i] +" ");
      }
      sc.close();
   }
}
```

```java
package module;
import java.util.Scanner;
public class SelectionSortBug {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        for (int i = 0; i < n - 1; i++) {
            int min = i;
            for (int j = i + 1; j < n; j++) {
                if (arr[j] < arr[min]) {
                    min = j;
                }
            }
            int temp = arr[i];
            arr[i] = arr[min];
            arr[min] = temp;
        }
        for (int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }
        sc.close();
    }
}
```

@ Javadoc Declaration Console

```
<terminated> SelectionSortBug [Java Application] C:\Users\mahi\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-06:
5
9 2 8 1 5
1 2 5 8 9
```